



Akash Chauhan

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BSc (Hons.) Data Science and AI

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GitHub | Portfolio

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EDUCATION

Degree Discipline	Institute/Board	CGPA/Percentage	Year
BSc (Hons.) Data Science and AI	Indian Institute of Technology, Guwahati	8.45 (Current)	2023-Present
Diploma in Electrical Engineering	Board of Technical Education, U.P.	First Division	2021-2024
Senior Secondary	Alpine Public School, Khurja (CBSE)	72.6%	2021
Secondary	Alpine Public School, Khurja (CBSE)	80.8%	2019

EXPERIENCE

- Sensing Perception & Intelligence (SPIN) Lab, IIT Guwahati | Research Intern** Nov 2025 – April 2026
Natural Scene Classification using Machine Learning and Computer Vision GitHub
 - Extended a natural scene classification project by scaling datasets from **100→2000 images** (binary classification) and **600→6000 images** (Intel multi-class classification) for more robust experimentation and evaluation.
 - Designed feature-engineering pipelines using statistical, texture, and colour descriptors including Entropy, Edge Density, GLCM features (Contrast, Homogeneity, Energy, Correlation), and HSV colour features.
 - Performed exploratory data analysis and feature-space interpretation using statistical significance testing (Mann–Whitney U, Kruskal–Wallis), correlation analysis, PCA, and t-SNE visualizations.
 - Implemented and compared Logistic Regression, SVM, and Random Forest classifiers using 5-Fold Cross Validation, feature importance analysis, and misclassification analysis.
 - Improved Intel multi-class classification accuracy from **52.5% to 64.7%** and achieved **88.5% accuracy** with **89.05% F1-score** on binary classification using Random Forest.
 - Tools/Tech:** Python, Numpy, OpenCV, Scikit-learn, Machine Learning, Computer Vision, Feature Engineering, Statistical Analysis, PCA, t-SNE

PROJECTS

- Fake News Detection System** April 2026
Python, NLP, Scikit-learn, NLTK, TF-IDF, Streamlit GitHub | Live Demo
 - Built an end-to-end fake news detector trained on 6000+ real-world news articles
 - Applied NLP techniques (text cleaning, stopword removal, TF-IDF vectorization) for feature extraction
 - Trained and compared 3 ML models (Logistic Regression, Naive Bayes, Random Forest)
 - Achieved 94%+ accuracy and deployed the model as a live Streamlit application
 - Deployed as a live interactive web app using Streamlit Cloud
- E-commerce Sales Prediction** April 2026
Python, Pandas, Scikit-learn GitHub
 - Built a machine learning model to predict sales using real-world transactional data
 - Performed data cleaning, feature engineering, and exploratory data analysis (EDA)
 - Implemented Linear Regression and Random Forest models; achieved improved performance using Random Forest for non-linear patterns
 - Identified key insights such as seasonal sales trends and country-wise revenue distribution
- Predicting Mechanical Properties of Lattice Structures using ML** March 2026
Python, Scikit-learn, Matplotlib GitHub
 - Developed a machine learning model to predict stiffness of lattice structures using geometric parameters such as density, thickness, and cell size
 - Generated a synthetic dataset to simulate real-world material behavior and trained Linear Regression and Random Forest models
 - Evaluated model performance using Mean Squared Error (MSE) and analyzed feature importance to identify key factors influencing stiffness
 - Demonstrated application of ML in mechanical metamaterials and biomedical design

TECHNICAL SKILLS

- Programming :** Python, SQL, R, Java, C
- ML & Data Science :** Scikit-learn, Pandas, NumPy, EDA, Feature Engineering, Statistical Analysis
- Visualization & BI :** Matplotlib, Power BI, Excel
- Tools :** Git, GitHub, VS Code, Jupyter Notebook, Google Colab

KEY COURSES TAKEN

- **Mathematics & Statistics** : Linear Algebra, Calculus, Optimization, Probability, Statistics
- **Data Science & AI** : Machine Learning, Deep Learning, Computer Vision, Time Series Forecasting, Recommender Systems, Data Mining, Data Visualization, RDBMS

CERTIFICATES

- **Deloitte Australia – Data Analytics Job Simulation**
Forage Feb 2026
– Completed practical tasks involving data analysis and business insights for a simulated industry scenario
- **Workshop on AI with Machine Learning**
IIT Roorkee Feb 2025
– Received hands-on exposure to machine learning concepts, algorithms, and real-world applications
- **Trustworthy Generative AI**
Vanderbilt University (Coursera) May 2024
– Studied principles of responsible AI development, safety, fairness, and ethical deployment of generative models
- **Generative AI for Executives and Business Leaders**
IBM May 2024
– Gained understanding of business applications, strategic use cases, and limitations of Generative AI
- **Ethical Hacking – AI Chatbots**
Udemy May 2024
– Explored security vulnerabilities, attack vectors, and defense strategies in AI-based chatbot systems
- **Introduction to R Programming**
Great Learning May 2024
– Learned fundamentals of R for data analysis, visualization, and statistical computing

ACHIEVEMENTS

- **Academic Excellence**, Achieved a CPI of **9.5** in the First Trimester at IIT Guwahati 2023
- **Grade AA (10/10)**, Awarded top grade in C Programming, AI Basics, and R Programming courses 2023–2024
- **Consistent Performance**, Maintaining a strong overall CPI of **8.45** across all trimesters Ongoing
- **Diploma with Distinction**, Completed 3-Year Diploma in Electrical Engineering with First Division 2024

ADDITIONAL INFORMATION

- **Languages** : Hindi (Native), English (Professional Proficiency)
- **Interests** : Data-Driven Problem Solving, Data Analytics, Emerging Technologies, Business Analytics